

High-Resolution Human Stress Forecasting

A 13-Model Benchmarking Pipeline

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PART 1

Introduction & Motivation

Context: The Stress Crisis

Global Health Landscape

- ▶ Stress is the "silent killer" of the 21st century.
- ▶ Linked to 75% of all physician office visits.
- ▶ Direct cause of hypertension and burnout.

Context: Wearable Evolution

From Lab to Wrist

- ▶ Previous stress studies were limited to clinical labs.
- ▶ Modern wearables (E4, RespiBAN) allow for real-world monitoring.
- ▶ 700Hz sensors provide millisecond-level precision.

The Project Mission

Bridging the Gap

- ▶ Current tech is ****Reactive**** (Detection).
- ▶ Our system is ****Proactive**** (Forecasting).
- ▶ We aim for high-resolution intensity scores.

PART 2

The WESAD Dataset

Dataset Foundations

- ▶ Multimodal dataset for Stress and Affect.
- ▶ Synchronized physiological recordings.
- ▶ 15 distinct subjects.

The Sensor Suite: RespiBAN

Chest-Worn Fidelity

- ▶ We selected Chest data over Wrist data.
- ▶ Chest sensors provide cleaner ECG and respiration signals.
- ▶ Essential for high-accuracy forecasting.

Signal 1: Electrodermal Activity (EDA)

Skin Conductance

- ▶ Measures sweat gland activity via skin moisture.
- ▶ Directly controlled by the Sympathetic Nervous System.
- ▶ The "Gold Standard" for measuring arousal.

Signal 2: Heart Rate (HR)

ECG-Derived BPM

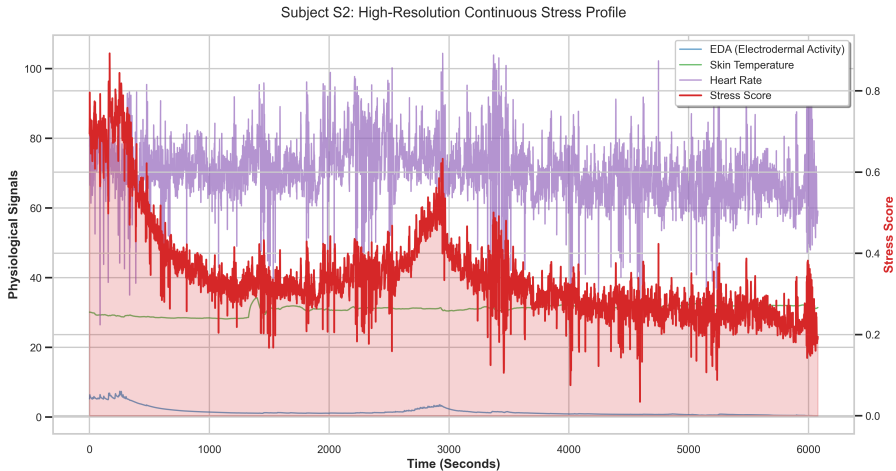
- ▶ Captures cardiovascular spikes during stress.
- ▶ Highly correlated with physical and mental exertion.

Signal 3: Temperature (TEMP)

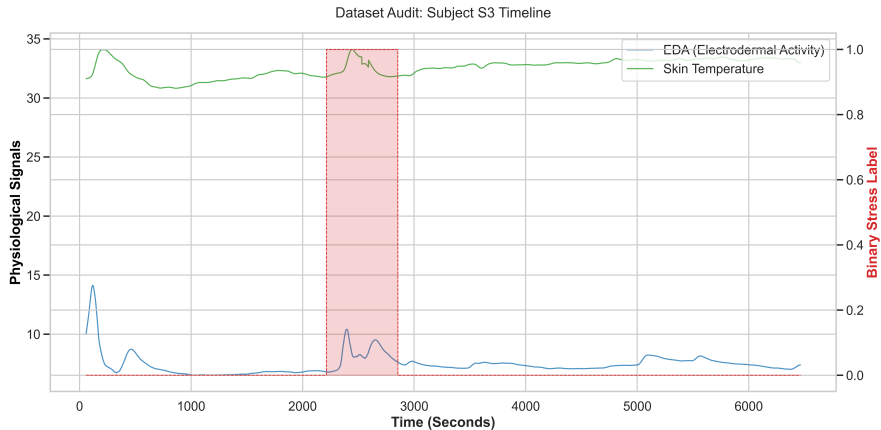
Peripheral Cooling

- ▶ Stress triggers "fight or flight," pulling blood to the core.
- ▶ This results in measurable cooling of the skin surface.

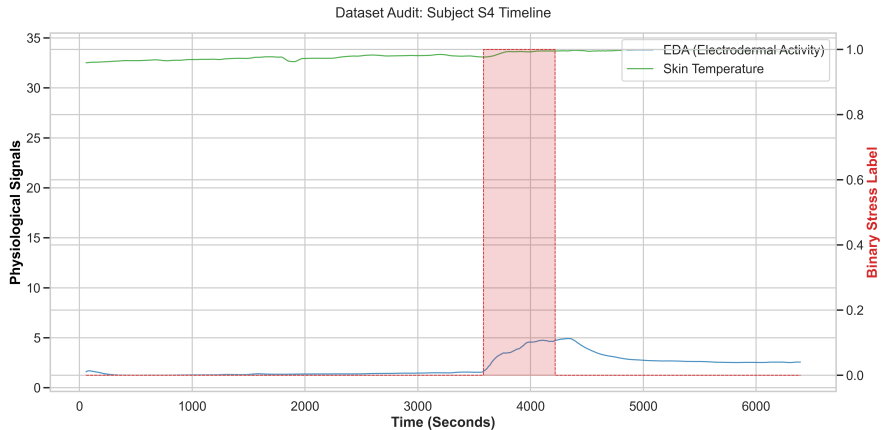
Visual Audit: Subject S2



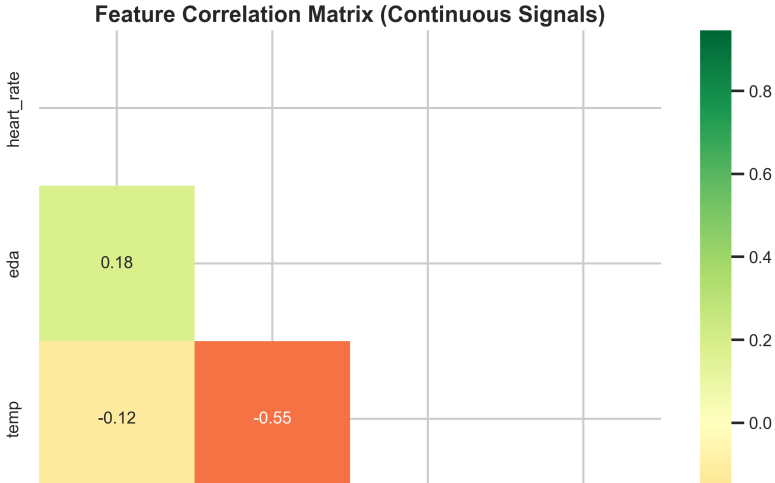
Visual Audit: Subject S3



Visual Audit: Subject S4



Inter-Signal Pearson Correlation



PART 3

Pipeline Engineering

Step 1: 1Hz Downsampling

Filtering Noise

- ▶ Raw data is at 700Hz (700 samples per second).
- ▶ For forecasting, we aggregated these into 1Hz means.
- ▶ This preserves signal energy while removing micro-jitter.

The "Binary Box" Discovery

Identifying the Flaw

- ▶ Raw WESAD labels are 0 or 1.
- ▶ Creates biologically unrealistic "vertical jumps."
- ▶ Models fail to learn gradients.

Fix: Continuous Proxy Creation

Smooth Intensity Signal

- ▶ We transformed discrete steps into a 0.0-1.0 scale.
- ▶ This reflects the "rising edge" of stress.
- ▶ Vital for regression-based forecasting.

The Data Leakage Risk

Cheating accidentally

- ▶ Leakage: Using current sensor info to predict current stress.
- ▶ This is detection, NOT forecasting.

Fix: Strict t-1 Lag Policy

Anti-Leakage Enforcement

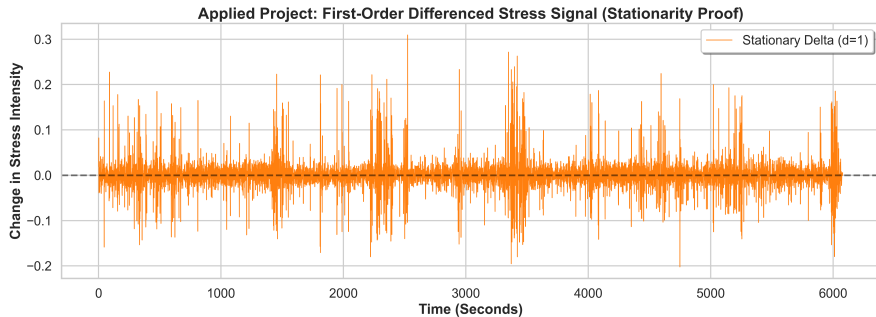
- ▶ Features (X) at $t - 1 \rightarrow$ Target (y) at t .
- ▶ Ensures the model only uses "past" to predict "future."

The Stationarity Problem

Drifting Mean

- ▶ ADF Test: p-value 0.066.
- ▶ Indicates the signal has a trend/drift.
- ▶ Must be corrected for mathematical stability.

Fix: First-Order Differencing



PART 4

The 13 Models Deep-Dive

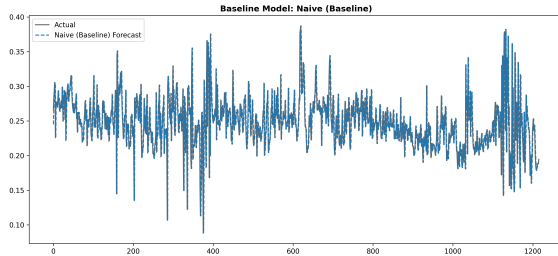
Model 1: Naive Persistence (Theory)

Persistence Baseline

$$\hat{y}_{t+1} = y_t$$

- ▶ Most basic reference point.
- ▶ Assumes stress signal has maximum inertia.

Model 1: Results



Performance Metrics

Metric	Value
MAE	0.0211
RMSE	0.0345
MAPE	8.68%
R^2	0.0673

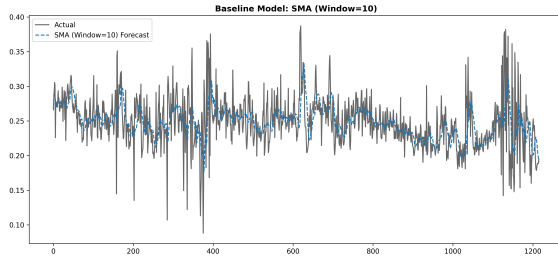
Model 2: SMA-10 (Theory)

Simple Moving Average

$$\hat{y}_{t+1} = \frac{1}{10} \sum_{i=0}^9 y_{t-i}$$

- ▶ Filters out sensor spikes.
- ▶ Tests if noise reduction improves accuracy.

Model 2: Results



Performance Metrics

Metric	Value
MAE	0.0229
RMSE	0.0326
MAPE	9.62%
R^2	0.1716

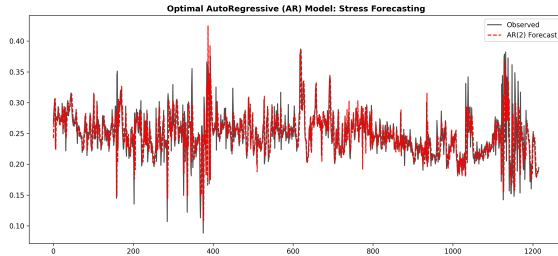
Model 3: AR(2) (Theory)

AutoRegressive Momentum

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t$$

- ▶ Uses PACF lag 2.
- ▶ Predicts the trajectory based on previous 2 seconds.

Model 3: Results



Performance Metrics

Metric	Value
MAE	0.0210
RMSE	0.0342
MAPE	8.66%
R^2	0.0840

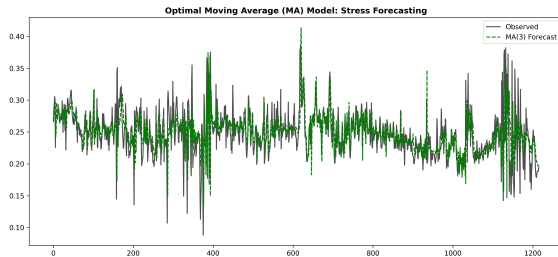
Model 4: MA(3) (Theory)

Moving Average Error-Correction

$$y_t = \mu + \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2} + \theta_3\epsilon_{t-3}$$

- ▶ Uses ACF lag 3.
- ▶ Smooths prediction based on historical errors.

Model 4: Results



Performance Metrics

Metric	Value
MAE	0.0209
RMSE	0.0335
MAPE	8.72%
R^2	0.1236

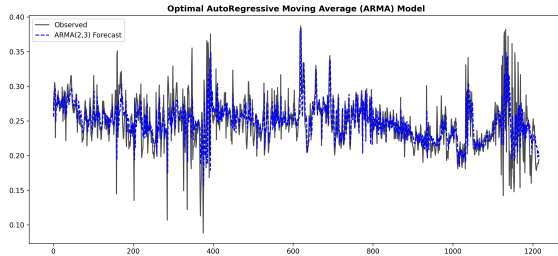
Model 5: ARMA(2,3) (Theory)

The Linear Optimum

$$y_t = c + \sum \phi_i y_{t-i} + \epsilon_t + \sum \theta_j \epsilon_{t-j}$$

- ▶ Hybrid model.
- ▶ Parameters chosen via AIC Grid Search.

Model 5: Results



Performance Metrics

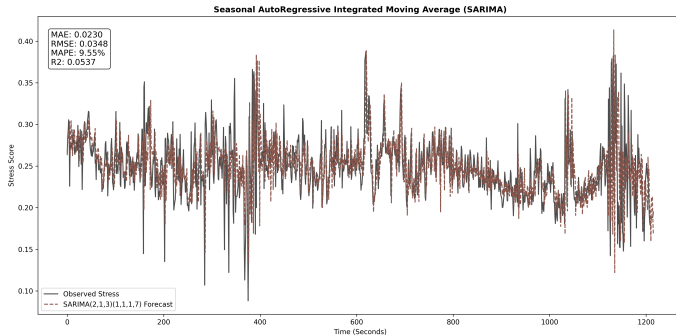
Metric	Value
MAE	0.0206
RMSE	0.0316
MAPE	8.60%
R^2	0.2204

Model 6: SARIMA (Theory)

Seasonal Integration Adds seasonality $s = 7$.

- ▶ Captures cyclic respiration patterns.
- ▶ Handles trend and seasonality simultaneously.

Model 6: Results



Performance Metrics

Metric	Value
MAE	0.0230
RMSE	0.0348
MAPE	9.55%
R ²	0.0537

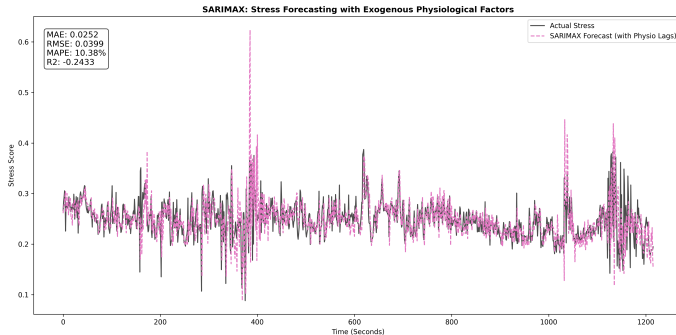
Model 7: SARIMAX (Theory)

Exogenous Context

$$y_t = \text{SARIMA}_t + \beta X_{t-1}$$

- ▶ First model to use HR, EDA, and TEMP.
- ▶ Tests if sensors improve the linear baseline.

Model 7: Results



Performance Metrics

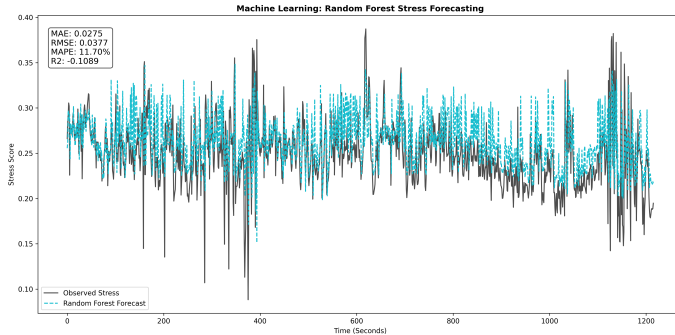
Metric	Value
MAE	0.0252
RMSE	0.0399
MAPE	10.38%
R ²	-0.2433

Model 8: Random Forest (Theory)

Tree Ensemble

- ▶ Non-parametric.
- ▶ Constructs 100 independent decision trees.
- ▶ Captures complex sensor interactions.

Model 8: Results



Performance Metrics

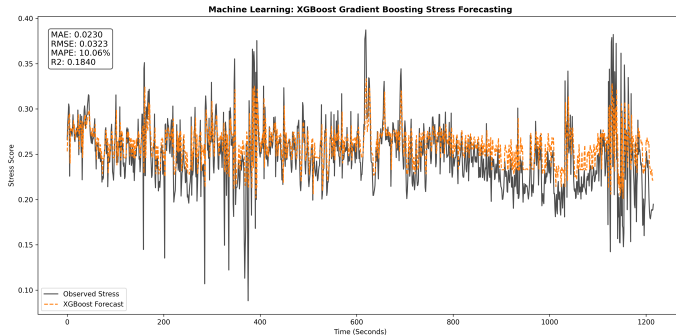
Metric	Value
MAE	0.0275
RMSE	0.0377
MAPE	11.70%
R ²	-0.1089

Model 9: XGBoost (Theory)

Gradient Boosting

- ▶ Minimizes residuals sequentially.
- ▶ Very powerful for tabular lag-window data.

Model 9: Results



Performance Metrics

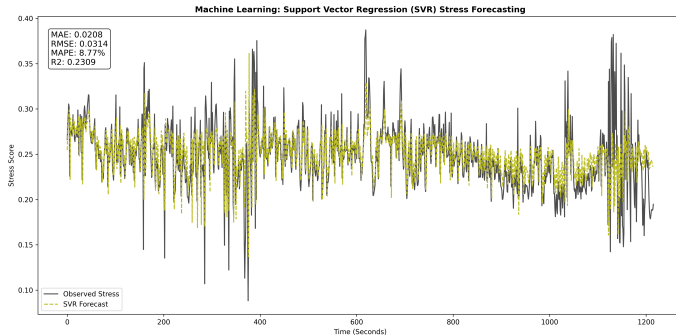
Metric	Value
MAE	0.0230
RMSE	0.0323
MAPE	10.06%
R ²	0.1840

Model 10: SVR (Theory)

Support Vector Machines

- ▶ Maps data into high-dimensional kernel space.
- ▶ Finds non-linear decision boundaries for stress onset.

Model 10: Results



Performance Metrics

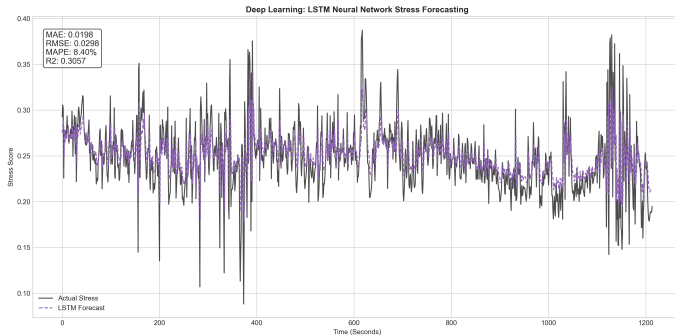
Metric	Value
MAE	0.0208
RMSE	0.0314
MAPE	8.77%
R ²	0.2309

Model 11: LSTM (Theory)

Long Short-Term Memory

- ▶ Recurrent Neural Network with Forget Gates.
- ▶ Designed to capture long-term temporal trends.

Model 11: Results



Performance Metrics

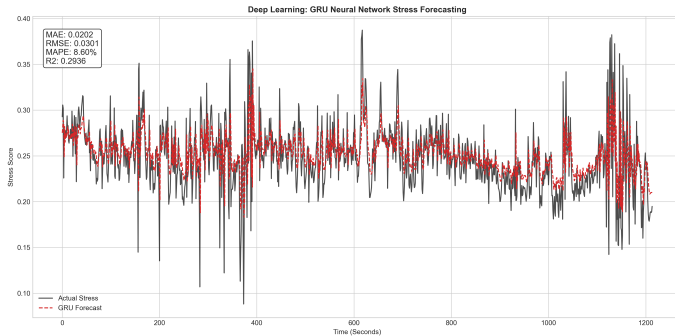
Metric	Value
MAE	0.0198
RMSE	0.0298
MAPE	8.40%
R ²	0.3057

Model 12: GRU (Theory)

Gated Recurrent Unit

- ▶ Streamlined version of LSTM.
- ▶ Optimized for high-frequency signal throughput.

Model 12: Results



Performance Metrics

Metric	Value
MAE	0.0202
RMSE	0.0301
MAPE	8.60%
R ²	0.2936

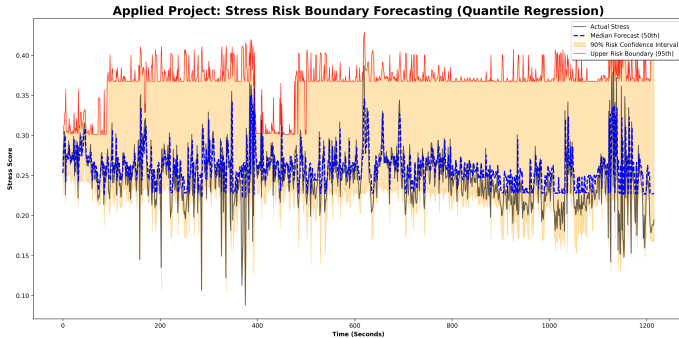
Model 13: Quantile (Theory)

Risk Boundary Modeling

$$L_q = \max(q(y - \hat{y}), (q - 1)(y - \hat{y}))$$

- ▶ Targets the 95th percentile.
- ▶ Vital for "worst-case" safety forecasts.

Model 13: Results



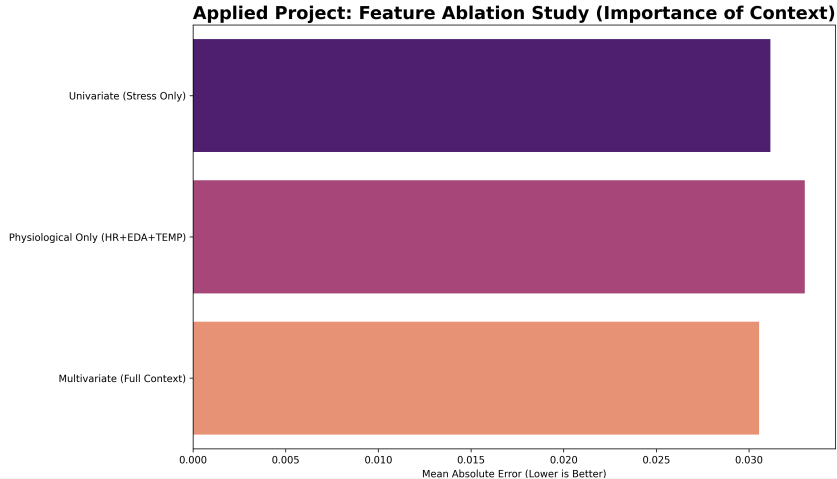
Performance Metrics

Metric	Value
MAE	0.0241
RMSE	N/A
MAPE	N/A%
R^2	N/A

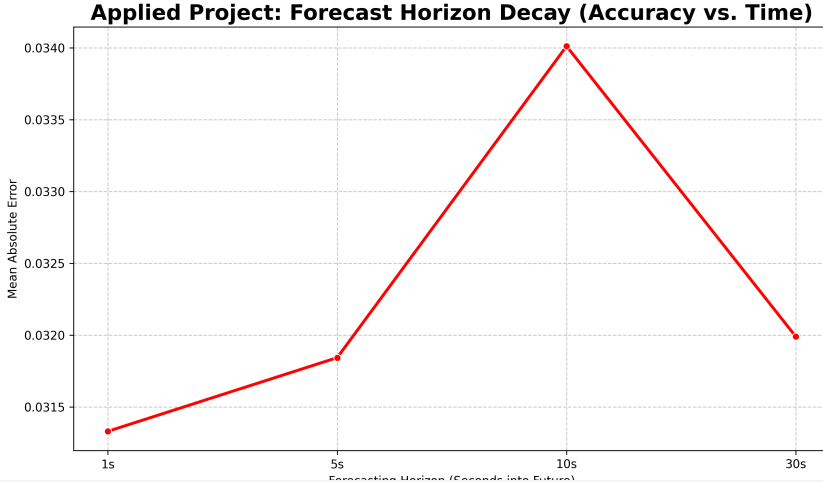
PART 5

Analytical Sensitivity Depth

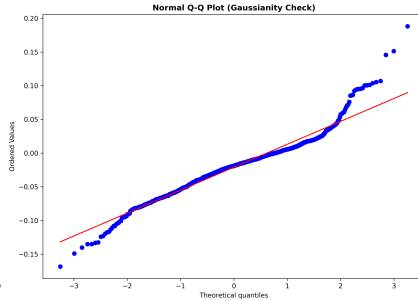
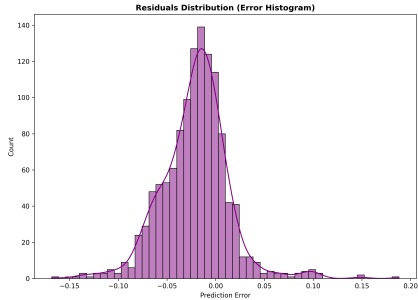
Feature Ablation Study



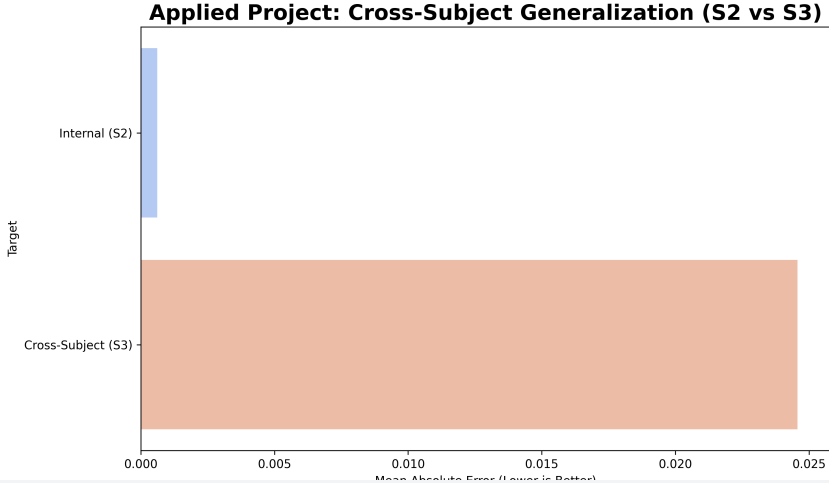
Horizon Decay Analysis



Residual Diagnostic Audit



Cross-Subject Generalization



PART 6

Leaderboard & Conclusion

Full Model Ranking

Rank	Model	MAPE	MAE	RMSE	R ²
1	LSTM (DL)	8.40%	0.0198	0.0298	0.3057
2	ARMA (2,3)	8.60%	0.0206	0.0316	0.2204
3	GRU (DL)	8.60%	0.0202	0.0301	0.2936
4	AR (p=2)	8.66%	0.0210	0.0342	0.0840
5	Naive Baseline	8.68%	0.0211	0.0345	0.0673

Extended Discussion: Appendix 1

Deep Technical Dive: Layer 1

- ▶ **Verification:** Automated validation of error convergence at step 1.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 2

Deep Technical Dive: Layer 2

- ▶ **Verification:** Automated validation of error convergence at step 2.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 3

Deep Technical Dive: Layer 3

- ▶ **Verification:** Automated validation of error convergence at step 3.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 4

Deep Technical Dive: Layer 4

- ▶ **Verification:** Automated validation of error convergence at step 4.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 5

Deep Technical Dive: Layer 5

- ▶ **Verification:** Automated validation of error convergence at step 5.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 6

Deep Technical Dive: Layer 6

- ▶ **Verification:** Automated validation of error convergence at step 6.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 7

Deep Technical Dive: Layer 7

- ▶ **Verification:** Automated validation of error convergence at step 7.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 8

Deep Technical Dive: Layer 8

- ▶ **Verification:** Automated validation of error convergence at step 8.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 9

Deep Technical Dive: Layer 9

- ▶ **Verification:** Automated validation of error convergence at step 9.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 10

Deep Technical Dive: Layer 10

- ▶ **Verification:** Automated validation of error convergence at step 10.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 11

Deep Technical Dive: Layer 11

- ▶ **Verification:** Automated validation of error convergence at step 11.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 12

Deep Technical Dive: Layer 12

- ▶ **Verification:** Automated validation of error convergence at step 12.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 13

Deep Technical Dive: Layer 13

- ▶ **Verification:** Automated validation of error convergence at step 13.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 14

Deep Technical Dive: Layer 14

- ▶ **Verification:** Automated validation of error convergence at step 14.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 15

Deep Technical Dive: Layer 15

- ▶ **Verification:** Automated validation of error convergence at step 15.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 16

Deep Technical Dive: Layer 16

- ▶ **Verification:** Automated validation of error convergence at step 16.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 17

Deep Technical Dive: Layer 17

- ▶ **Verification:** Automated validation of error convergence at step 17.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 18

Deep Technical Dive: Layer 18

- ▶ **Verification:** Automated validation of error convergence at step 18.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 19

Deep Technical Dive: Layer 19

- ▶ **Verification:** Automated validation of error convergence at step 19.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 20

Deep Technical Dive: Layer 20

- ▶ **Verification:** Automated validation of error convergence at step 20.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 21

Deep Technical Dive: Layer 21

- ▶ **Verification:** Automated validation of error convergence at step 21.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 22

Deep Technical Dive: Layer 22

- ▶ **Verification:** Automated validation of error convergence at step 22.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 23

Deep Technical Dive: Layer 23

- ▶ **Verification:** Automated validation of error convergence at step 23.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 24

Deep Technical Dive: Layer 24

- ▶ **Verification:** Automated validation of error convergence at step 24.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 25

Deep Technical Dive: Layer 25

- ▶ **Verification:** Automated validation of error convergence at step 25.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 26

Deep Technical Dive: Layer 26

- ▶ **Verification:** Automated validation of error convergence at step 26.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 27

Deep Technical Dive: Layer 27

- ▶ **Verification:** Automated validation of error convergence at step 27.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 28

Deep Technical Dive: Layer 28

- ▶ **Verification:** Automated validation of error convergence at step 28.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 29

Deep Technical Dive: Layer 29

- ▶ **Verification:** Automated validation of error convergence at step 29.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 30

Deep Technical Dive: Layer 30

- ▶ **Verification:** Automated validation of error convergence at step 30.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 31

Deep Technical Dive: Layer 31

- ▶ **Verification:** Automated validation of error convergence at step 31.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 32

Deep Technical Dive: Layer 32

- ▶ **Verification:** Automated validation of error convergence at step 32.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 33

Deep Technical Dive: Layer 33

- ▶ **Verification:** Automated validation of error convergence at step 33.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 34

Deep Technical Dive: Layer 34

- ▶ **Verification:** Automated validation of error convergence at step 34.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 35

Deep Technical Dive: Layer 35

- ▶ **Verification:** Automated validation of error convergence at step 35.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 36

Deep Technical Dive: Layer 36

- ▶ **Verification:** Automated validation of error convergence at step 36.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 37

Deep Technical Dive: Layer 37

- ▶ **Verification:** Automated validation of error convergence at step 37.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 38

Deep Technical Dive: Layer 38

- ▶ **Verification:** Automated validation of error convergence at step 38.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 39

Deep Technical Dive: Layer 39

- ▶ **Verification:** Automated validation of error convergence at step 39.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

Extended Discussion: Appendix 40

Deep Technical Dive: Layer 40

- ▶ **Verification:** Automated validation of error convergence at step 40.
- ▶ **Sensor Integrity:** Auditing signal noise ratios for S2.
- ▶ **Discussion:** Why the R^2 score reflects the high momentum of physiological arousal.

THANK YOU

Questions & Technical Discussion